OpenMV® PS5520 5MP HDR Camera Module

Datasheet

SKU: OMV-MOD-HDR-V1

Description

The PS5520 is a highly integrated CMOS image sensor that outputs 2592x1944 (5 MP) pixels with a rolling shutter readout. It features FinePixel and HDR sensor technology for excellent image quality and single-shot high dynamic range synthesized image output. VGA (640x480), HD (1280x720), and FHD (1920x1080) are operational at 30 FPS with HDR support.

Target Areas

High-resolution imaging, HDR applications, outdoor vision, security, industrial inspection, quality control.

Key Features

- **5 MP resolution** — 2592x1944 pixel output with rolling shutter.
- **HDR imaging** — FinePixel + HDR technology for single-shot high dynamic range.
- **30 FPS at FHD** — VGA, HD, and FHD modes at 30 FPS with HDR.
- **N6 compatible** — Designed for the OpenMV N6.

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1 Specifications

1.1 Sensor Specifications

Component	Details
Sensor	PS5520 CMOS image sensor.
Resolution	2592x1944 (5 MP).
Shutter	Rolling shutter.
Technology	FinePixel + HDR for single-shot high dynamic range.

1.2 Operating Modes

Component	Details
VGA	640x480 at 30 FPS with HDR.
HD	1280x720 at 30 FPS with HDR.
FHD	1920x1080 at 30 FPS with HDR.
QWXGA2 (planned)	2952x1944 — in development, expected by end of summer 2026.

1.3 Compatibility

Component	Details
Supported Cameras	OpenMV N6 only.

2 General Info

Dimension	Value
Weight	30 g

3 Company Information

Field	Details
Website	https://openmv.io

4 Reference Documentation

Reference	Link / Document
Product Page	https://openmv.io/products/ps5520-5mp-hdr-camera-module
OpenMV N6	https://openmv.io/products/openmv-n6
Documentation	https://docs.openmv.io

5 Revision History

Date	Revision	Changes
2026-04-04	Rev 1	Initial public release.